Gerasimos (Makis) Lampouras

Chief Scientific Officer at C Proof, London

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https://glampouras.github.io/



Short Bio

Gerasimos (Makis) Lampouras is the Chief Scientific Officer at C Proof, London. Prior to this, he was a Principal Research Scientist and served as the Team Leader of the NLP (Natural Language Processing) group in Huawei Noah's Ark Lab, London.

Areas of interest generally fall somewhere between Natural Language Processing and Machine Learning, with his latest efforts focusing on the information seeking and synthesis capabilities of LLM-based Agents and on neuro-symbolic approaches for math and code reasoning. Briefly, his previous work has focused on LLM-based code synthesis and understanding, Natural Language Generation, Dialogue Systems, and overall Machine Learning applications for Natural Language Processing.

Earlier, he was a Research Associate at the University of Cambridge, University of Sheffield and University College London. He received his MSc and PhD from the Athens University of Economics and Business.

Employment History

2026 – · · · ·
Chief Scientific Officer at C Proof, London.
Leading research on automatic verification for code agents, building the runtime (Chelis; https://chelis.ch/) that ensures AI-generated code is correct and safe prior to execution. Spearheading technical strategy and domain applications, starting with quantitative finance (C Note; https://cproof.ai/c-note).

2024 – 2026
Team Leader of the NLP (Natural Language Processing) Group at Huawei Technologies Research & Development (UK) Ltd, London.
Leading a group of ∼10 Senior Research Scientists and Engineers researching how to address current NLP and Large Language Models challenges. Most recently focusing on the information seeking and synthesis capabilities of LLM-based Agents and on neuro-symbolic approaches for math and code reasoning, including supervised finetuning, reinforcement learning, self-learning, efficiency, and related tasks.

2023 – 2026
Principal Research Scientist in NLP at Huawei Technologies Research & Development (UK) Ltd, London.
Research management and development of Code Language Models, including the full development pipeline of a foundational Code Language Model (data collection, pretraining, finetuning).

2019 – 2023
Senior Research Scientist in NLP at Huawei Technologies Research & Development (UK) Ltd, London.
Research and development focused on multilingual Natural Language Generation, evaluation, and Dialogue Systems.

2018 – 2019
Research Associate at University of Cambridge.
EPSRC eNeMILP project: Research on non-monotonic incremental prediction of structured outputs.

Employment History (continued)

2015 – 2018 **Research Associate** at University of Sheffield.
EPSRC DILiGENt project: Research on domain-independent Natural Language Generation through imitating generation policies from unaligned corpora.
EPSRC eNeMILP project: Research on non-monotonic incremental prediction of structured outputs.

2015 **Research Associate** at University College London.
EPSRC DILiGENt project: Research on domain-independent Natural Language Generation through imitating generation policies from unaligned corpora.

2014 **Research Assistant** at Athens University of Economics and Business.
BioASQ: A Challenge on Large-Scale Biomedical Semantic Indexing and QA

2011 – 2012 **Research Assistant** at Athens University of Economics and Business.
Researching a Linear Programming approach to multi-document text Summarization and Natural Language Generation from ontologies.

2007 – 2009 **Research Assistant** at Athens University of Economics and Business.
INDIGO: Interaction with Personality and Dialogue Enabled Robots

Education

2009 – 2015 **Ph.D. in Artificial Intelligence, Natural Language Processing** at Athens University of Economics and Business.
Thesis title: *Natural language interaction with Semantic Web Ontologies.*

2006 – 2008 **M.Sc. in Computer Science** at Athens University of Economics and Business.
Dissertation title: *Methods to automatically detect definitions in document collections.*

2002 – 2006 **Bachelor's degree Informatics** at Athens University of Economics and Business.
Dissertation title: *Revision and larger-scale experimental evaluation of a method to handle definition questions in question answering systems.*

Selected Research Publications

M. Gritta, D. Paul, X. Li, L. Shang, J. Wang, and G. Lampouras, "**Process evaluation for agentic systems**," ***Proceedings of EACL - Findings***, 2026.

D. Paul, D. Murphy, M. Gritta, R. Cardenas, V. Prokhorov, L. S. Bolliger, A. Toker, R. Miles, A.-M. Oncescu, J. A. Sivakumar, P. Borchert, I. Elezi, M. Zhang, K. Y. Lee, G. Zhang, J. Wang, and G. Lampouras, "**A benchmark for deep information synthesis**," ***Proceedings of ICLR***, 2026.

M. Zhang, P. Borchert, M. Gritta, and G. Lampouras, "**DRIFT: Decompose, retrieve, illustrate, then formalize theorems**," ***Proceedings of ICLR***, 2026.

F. Christopoulou, R. Cardenas, G. Lampouras, H. Bou-Ammar, and J. Wang, "**SparsePO: Controlling preference alignment of LLMs via sparse token masks**," ***Proceedings of EMNLP - Findings***, 2025.

M. Gritta, H. Xue, and G. Lampouras, "**DReSD: Dense retrieval for speculative decoding**," ***Proceedings of ACL - Findings***, 2025.

Y. Li, P. Borchert, and G. Lampouras, "**TopoAlign: A framework for aligning code to math via topological decomposition**," ***arXiv pre-print***, 2025.

J. A. Sivakumar, P. Borchert, R. Cardenas, and G. Lampouras, "**Conjecturing: An overlooked step in formal mathematical reasoning**," ***arXiv pre-print***, 2025.

F. Christopoulou, G. Zhang, and G. Lampouras, "**Text-to-code generation with modality-relative pre-training**," ***Proceedings of EACL***, 2024.

- Z. Fountas, M. A. Benfeghoul, A. Oomerjee, F. Christopoulou, G. Lampouras, H. Bou-Ammar, and J. Wang, "**Human-like episodic memory for infinite context LLMs**," ***Proceedings of ICLR***, 2024.
- M. Gritta, G. Lampouras, and I. Iacobacci, "**HumanRankEval: Automatic evaluation of LMs as conversational assistants**," ***Proceedings of NAACL***, 2024.
- M. Zimmer, M. Gritta, G. Lampouras, H. B. Ammar, and J. Wang, "**Mixture of attentions for speculative decoding**," ***Proceedings of ICLR***, 2024.
- P. Chen and G. Lampouras, "**Exploring data augmentation for code generation tasks**," ***Proceedings of EACL***, 2023.
- P. J. Gorinski, M. Zimmer, G. Lampouras, D. G. X. Deik, and I. Iacobacci, "**Automatic unit test data generation and actor-critic reinforcement learning for code synthesis**," ***Proceedings of EMNLP***, 2023.
- F. Christopoulou, G. Lampouras, M. Gritta, G. Zhang, Y. Guo, Z. Li, Q. Zhang, M. Xiao, B. Shen, L. Li, et al., "**PanGu-Coder: Program synthesis with function-level language modeling**," ***arXiv pre-print***, 2022.
- F. Christopoulou, G. Lampouras, and I. Iacobacci, "**Training dynamics for curriculum learning: A study on monolingual and cross-lingual NLU**," ***Proceedings of EMNLP***, 2022.
- Y. Feng, G. Lampouras, and I. Iacobacci, "**Topic-aware response generation in task-oriented dialogue with unstructured knowledge access**," ***Proceedings of EMNLP***, 2022.
- G. Zhou, G. Lampouras, and I. Iacobacci, "**Hierarchical recurrent aggregative generation for few-shot NLG**," ***Proceedings of ACL***, 2022.
- M. Gritta, G. Lampouras, and I. Iacobacci, "**Conversation graph: Data augmentation, training, and evaluation for non-deterministic dialogue management**," ***TACL***, vol. 9, pp. 36–52, 2021.
- G. Zhou and G. Lampouras, "**Generalising multilingual concept-to-text NLG with language agnostic delexicalisation**," ***Proceedings of ACL***, 2021.
- G. Zhou and G. Lampouras, "**Informed sampling for diversity in concept-to-text NLG**," ***Proceedings of EMNLP***, 2021.
- G. Lampouras and A. Vlachos, "**Imitation learning for language generation from unaligned data**," ***Proceedings of COLING***, 2016.

The full list of publications is available at `https://glampouras.github.io/`.

Selected Open-Source Projects

- **Chelis**: the verified tensor language for agent-written code, developed at C Proof. Shapes, precision and ownership are checked at compile time, and stated properties are proved with an SMT solver.
  `https://github.com/Chelis-Lang/chelis`
- **DeepSynth**: a benchmark of 120 real-world tasks testing whether AI agents can gather, combine, and reason over information from multiple sources (ICLR 2026).
  `https://github.com/agentdeepsynthesis/deepsynth-bench`
- **Formal Mathematical Reasoning**: bridging natural language mathematics and machine-verifiable proofs through autoformalisation; includes LeanFlow, a Python interface for Lean 4.
  `https://github.com/Formal-Math-Reasoning`

The full list of open-source projects is available at `https://glampouras.github.io/`.

Granted Patents

- P. J. Gorinski, M. Zimmer, G. Lampouras, D. G. X. Deik, and I. Iacobacci, "Apparatus and method for the training of a code synthesis model," 2024.
- G. Zhou and G. Lampouras, "Hierarchical language generation with recurrent aggregation," 2023.
- G. Zhou and G. Lampouras, "Device and method for generating language," 2021.

Event Organization

2024 Co-organizer of SCI-CHAT: Workshop on Simulation of Conversational Intelligence in Chat.
2022 Co-organizer of the 6th Structured Prediction in Natural Language Processing workshop.
2020 Co-organizer of the 4th Structured Prediction in Natural Language Processing workshop.
2019 Co-organizer of the 3rd Structured Prediction in Natural Language Processing workshop.

Chairing and Service

2023 – · · · · Acting as ACL ARR Area Chair.
2024 Acted as "Summarization" Area Chair for EACL 2024.
Acted as "Evaluation and Validation Methodologies" Area Chair at LREC-COLING 2024.
2023 Acted as "Generation & Summarization" Area Chair for EACL 2023.
2021 Acted as "Resources and Evaluation" Area Chair for EMNLP 2021.
2020 Acted as Sponsorship Chair for INLG 2020.

Honors and Awards

2025 Huawei Team Gold Medal Award
2022 Huawei Innovation Pioneer Award
For PanGu-Coder: Program synthesis with function-level language modeling.
2020, 2023, 2025 Huawei Rising Star Awards
2010 – 2013 Research Funding Program: Heracleitus II
Ph.D. research fund co-financed by the European Union (European Social Fund – ESF) and Greek national funds through the Operational Program "Education and Lifelong Learning" of the National Strategic Reference Framework (NSRF).
2006 – 2007 MSc scholarships
Scholarships awarded in the first and second semester (3rd and 1st best position respectively) of the MSc program.